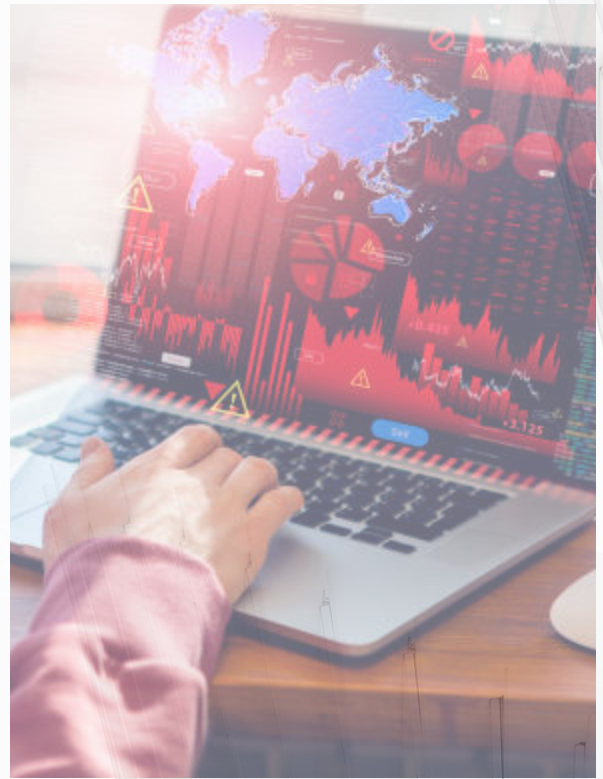


MSC DATA SCIENCE CAPSTONE PROJECTS 2022/23



The capstone is a collaborative project, providing students with the opportunity to study in depth a topic of specific interest. The topic is proposed by an industry partner and will normally relate to data-based issues faced in practice by private or public sector organisations. This will require the use of specific data sources and data science skills learnt on the programme.

The capstone project is conducted in partnership with a company partner and is jointly supervised by the LSE faculty and company partner collaborators. The capstone project partner proposes a data science research project, potentially provides access to data, and engages through participation in joint meetings.

The capstone project requires creative work in formulating research questions and hypotheses, identifying the most suited methodology, referring to research literature, and analysing data sources using a variety of data science tools.

**For more information, please contact
Dr. Marcos Barreto, m.e.barreto@lse.ac.uk**

STUDENT DELIVERABLES

Capstone projects run for 10 months (November to August). The students are expected to engage with recurrent meetings with their industry partners and LSE supervisors and work enough weekly hours to comply with the expected outcomes.

The students should attend two all-hands meetings (February and April) to present their work and collect feedback from peers and LSE staff. A draft report is submitted in May, and a final report is due in August. Any other deliverables (such as technical papers, software, etc) must be agreed among the students, LSE supervisor, and the industry partner.

EXAMPLE PROJECTS

Capstone projects require the use of data science methods and tools to solve problems in different areas, such as healthcare, finance, social sciences, media and entertainment, logistics, among others.

Here are some recent examples of topics:

- **Forecasting sea water transparency:** in partnership with a multinational technology conglomerate, this project has used meteorological, hydrographic, and other related signals, as well as historic observations from satellite images, to forecast water transparency and turbidity measures.
- **Reorder point optimisation:** in partnership with a global management consultancy company, this project has combined methods from Machine Learning and Operations Research to analyze time series data and determine the optimal safety stock level and reorder point.
- **Alternative data for thematic basket of assets:** joint with a world's leading financial service provider, this project has deployed a machine learning pipeline to analyse online trending topics and build thematic baskets of assets (such as stocks, currencies, commodities, bonds or cryptos) for monitoring their performance.
- **Recommendations of mental health interventions:** in partnership with a leading provider of digital solutions for mental health, this project has applied deep and reinforcement learning approaches to assess the efficacy of different intervention configurations (which intervention, in what sequence and at what time) in terms of prediction accuracy and estimation of impact.

<https://bit.ly/lsestats-capstone-projects>

MSC DATA SCIENCE CAPSTONE PROJECT

TIMELINE



MSC DATA SCIENCE CAPSTONE PROJECTS

FAQS

What makes a successful project?

A successful project is a combination of an interesting topic proposed by the partner, properly supported by data sources and research questions, and the skills and engagement presented by the students. A team of students with access to LSE's research resources and academic supervision can achieve sufficient depth and independence of analysis to kick-start further action and debate the proposal with the industry partner.

The most successful projects are those that set clear and realistic research goals and have sufficient and consistent technical guidance from the partner, particularly in facilitating access to the necessary data, indicating essential readings, discussing analytical methods, and validating outputs.

How can partners best support students?

The students are expected to take full responsibility for leading, planning and executing their project, as well as communicating with the partner and the LSE supervisor. The partner is responsible for providing a clear problem to be investigated, including interesting research questions, access to any proprietary data (apart from public datasets that can be used), and remain available and open to meet the students, hear their suggestions and address any issues.



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MSC DATA SCIENCE CAPSTONE PROJECTS

FAQS

How are capstones listed on CVs and LinkedIn?

Capstone projects can be noted on CVs and LinkedIn as part of the education, and not as work experience, as the capstone project can be understood as one of the courses the students take during their Master programme. Some partners may require a Non-Disclosure Agreement (NDA) preventing students to cite their names or any outputs from the capstone without explicit authorization.

What are the expectations regarding communications?

We ask each group of students to elect someone to be responsible for arranging regular communications with the partner and academic supervisor. This can be established during the kick-off meeting. These meetings are usually virtual but can accommodate any in-person activity when needed and feasible. Any other communication tools can be also discussed during the kick-off meeting.

LSE supervisors are expected to provide support to students mainly for the academic reports, up to two weeks after the end of the Summer Term. They can attend the regular meetings between partners and students and engage with any other specific aspects of the capstone project, if agreed.

Partners and LSE supervisors can meet to discuss specific points when necessary.



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MSC DATA SCIENCE CAPSTONE PROJECTS

FAQS

Are there any costs involved for partners?

There is no charge to companies for sponsoring a project. Students do not receive pay for completing work on the project. We do have a limited budget provided by LSE for these projects for things like travel, printing expenses, and conference fees.

We do not expect students to provide hardcopies of their slides and reports to faculty or partners. All deliverables are submitted electronically and there is no need for printed copies.

Whilst infrequent travel within London is to be expected, it would be appreciated if clients could cover the cost of more exceptional travel, such as trains to locations outside of London (for visiting partner premises, for instance). It is preferable, of course, that all meetings and interactions occur virtually.



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